Detecting AI-Generated Images in Practice: A Review and a Deployment Case Study

REDOANUZZAMAN
Sichuan University, Chengdu, China
2023521460126@stu.scu.edu.cn

***Abstract*—Passive detectors of AI-generated images report accuracies above 95% on curated benchmarks, yet fall to near chance on realistic images, unseen generators, and images that have been compressed or resized. This paper reviews the detection literature through the lens of deployment. We cover the main method families, the benchmarks used to rank them, the shortcuts that inflate reported numbers (container format, file size, resampling history), the gap between ranking quality and calibrated decisions, and the role of cryptographic provenance (C2PA) and watermarking. We then report a case study: an open, stateless web detector built from off-the-shelf parts, evaluated with the controls the review argues for. On a held-out set of 75 real photographs and 148 images from two diffusion generators, all normalized to a single container, averaging a Swinv2 classifier with the Community Forensics ViT in log-odds space raised ROC AUC from 0.919 to 0.956 (paired Δ = +0.037, 95% CI [+0.010, +0.068]). At a threshold tuned on separate data, it flagged 0 of 75 real photographs, against 5 of 75 for the single model, while catching 80% of fakes rather than 66%. A third candidate, whose scores barely moved under any degradation (max $|\Delta| = 0.007$), was rejected as a content-shortcut detector despite a competitive clean AUC. We distil the review and the case study into practical recommendations: report calibration beside AUC, include trivial baselines, normalize containers, verify provenance instead of sniffing it, and never let a detector score stand as proof.**

***Index Terms*—AI-generated image detection, image forensics, generalization, robustness, calibration, C2PA, content provenance, ensemble**

I. Introduction

Text-to-image models now produce photographs that people cannot tell from camera images. The fakes in the Chameleon dataset were all selected because human raters mistook them for real photos [1]. Detecting such images matters for misinformation, insurance and document fraud, academic integrity, and platform moderation. Two families of approach exist. *Active* methods embed a signal at generation time: a watermark [2], [3] or a signed provenance manifest [4]. They give strong guarantees, but only if the generator cooperates and the signal survives redistribution [5]. *Passive* methods infer synthesis from the pixels alone, and they are the only option for the large majority of images that carry neither signal.

The passive literature has grown quickly, from CNN artifact detectors [6] to frozen foundation-model features [7] and multi-expert systems [1]. So has evidence that its headline numbers do not transfer. A zero-shot audit of 16 open detectors on 12 datasets found no universal winner and detection rates of only 18–30% on recent commercial generators [8]. Popular training sets let a model separate the classes by JPEG quality or image size rather than by generation artifacts [9]. And the NTIRE 2026 challenge made robustness to chains of real-world distortions its primary metric [10].

This paper makes three contributions.

1) A **deployment-oriented review** of passive detection (Sections III–V). It organizes the method families, benchmarks and evaluation pitfalls around one question: which reported numbers survive contact with real images? It also covers the provenance signals that should sit beside any passive score.
2) A **case study** of an open, stateless detector (Section VI). It shows how the review's findings become engineering decisions: verdict bands instead of a 0.5 threshold, a region heatmap, verified rather than sniffed C2PA provenance, and a fetch path that cannot be turned against internal services.
3) A **controlled evaluation** (Section VII). It covers a robustness pilot, a paired comparison of three candidate second detectors on held-out data with container normalization and a trivial baseline, and a diagnostic that exposes a detector reading content rather than artifacts.

Section VIII turns both halves into recommendations and open problems.

II. Scope and Method of the Review

We started from a recent survey's taxonomy [11] and followed citations forward to work from 2023–2026 on generalization, robustness, dataset bias and calibration, plus the benchmark and challenge reports that define current practice [8], [10], [12]–[15]. Every quantitative claim taken from prior work was checked against the full text of the cited version. Where a figure differed between arXiv versions, we report the latest. Where two rows of a table used different training regimes, we report a like-for-like row. Preprints are cited as such. The review is narrative, not systematic: its aim is to explain which results matter for deployment, not to count papers.

III. Method Families

Table I summarizes the five passive families. Their core trade-off is between *what signal they read* and *what destroys it*.

TABLE I
PASSIVE DETECTION FAMILIES, THE SIGNAL EACH READS, AND WHAT TYPICALLY BREAKS IT.

Family	Representatives	Signal	Main weakness
Spatial artifact	CNNSpot [6], NPR [16], SSP [17], PatchCraft [18]	Up-sampling traces, local texture	Compression, resizing; unseen generators
Frequency	FreDect [19], Corvi et al. [20], STAL [21], S^2F-Net [22]	Spectral peaks, high-frequency tails	JPEG rewrites the band it reads
Foundation features	UnivFD [7], C2P-CLIP [23], SSAFE [24], PPM-CLIP [25]	Frozen CLIP/ViT embeddings + small head	Semantic shortcuts; adversarial fragility [26]
Reconstruction	DIRE [27], MIRROR [28]	Diffusion inversion error	Inference cost; generator-specific
Hybrid / explainable	AIDE [1], Mirage [29], AgentFoX [30]	Semantic + low-level experts; VLM rationales	Complexity; rationales unverified

TABLE II
BENCHMARKS REFERENCED IN THIS REVIEW.

Benchmark	Scale	What it tests
GenImage [12]	1.33M real / 1.35M fake, 8 generators	Cross-generator transfer; degraded images
Chameleon [1]	14,863 real / 11,170 fake	Human-deceptive realism
Community Forensics [13]	2.7M images, 4,803 generators	Generator diversity at scale
AI-GenBench [14]	continually extended	Temporal: train on past, test on future generators
NTIRE 2026 [10]	108,750 real / 185,750 fake, 42 generators	36 distortion types, chains of 1–5
ForensicHub [15]	unified codebase	Cross-domain forensics
Out-of-the-box audit [8]	16 methods, 12 datasets, 291 generators	Zero-shot transfer of released weights

A. Spatial artifact detectors

The field's starting point showed that a ResNet trained on one GAN generalizes surprisingly well to other CNN generators, provided training includes blur and JPEG augmentation [6]. Later work targets the traces of up-sampling more directly. NPR models the relationships between neighbouring pixels created by up-sampling layers [16]. SSP shows that the noise pattern of a single simple patch can suffice [17]. PatchCraft contrasts rich- and poor-texture regions and introduced the AIGCDetectBenchmark [18]. These detectors are cheap and often strong in-distribution, but the traces they read live at the scale that compression and resizing destroy first.

B. Frequency-domain detectors

Up-sampling leaves periodic peaks in the image spectrum that a classifier can exploit directly [19]. Diffusion models leave weaker but still measurable spectral fingerprints [20]. The known weakness is that JPEG quantization rewrites exactly the high-frequency band these methods read. Recent work treats this explicitly. STAL adds an auxiliary objective on the spectral tail and reports balanced accuracy on GenImage falling only from 98.62% to 96.24% between JPEG quality 100 and 60 [21]. S^2F-Net fuses spatial and spectral streams for cross-model transfer [22].

C. Foundation-model features

The current mainstream freezes a large pretrained encoder and trains a small head. UnivFD's linear probe on CLIP ViT-L/14 [7], [31] showed that such features generalize from GANs to diffusion models far better than detectors trained from scratch. The idea has since been refined with prompt learning [23], [25] and stronger frozen encoders [24]. The NTIRE 2026 winners fine-tuned DINOv3 backbones [10], [32]. The generalization is real but uneven. Re-evaluated on AIGCDetectBenchmark after ProGAN training, UnivFD scores 99.81% on ProGAN and 50.75% on DALL·E 2 [1], a coin flip on an unseen family. Semantic features also invite semantic shortcuts: a head can learn *what* generators tend to depict rather than *how* they render it. Section VII-D shows a deployed model that appears to do exactly this. CLIP-based detectors are also vulnerable to adversarial perturbation [26].

D. Reconstruction-based detectors

DIRE observes that images produced by a diffusion model are reconstructed more faithfully when inverted through a diffusion model than camera images are, and uses the reconstruction error as the feature [27]. MIRROR is a recent descendant [28]. The cost is one or more diffusion passes per image, and the signal depends on how close the inverting model is to the generator.

E. Hybrid and explainable detectors

AIDE combines a CLIP semantic expert with a low-level expert fed the highest- and lowest-frequency patches. It improves on the previous best, PatchCraft, by 3.5% on AIGCDetectBenchmark and 4.6% on GenImage [1]. A newer line uses vision-language models or LLM agents to name the artifact behind a decision [29], [30]. That makes outputs easier to read, but whether the stated rationale reflects the evidence the model actually used is largely unmeasured.

IV. BENCHMARKS AND EVALUATION PRACTICE

Table II lists the benchmarks this review draws on. Four lessons emerge from them.

A. The generalization gap is the rule

Chameleon collects fakes that human raters took for real photos. When trained on all of GenImage, CNNSpot, FreDect, PatchCraft and AIDE reach only 60.89%, 57.22%, 55.70% and 65.77% accuracy on it [1]. Chance is 50%. The largest zero-shot audit to date, covering 16 methods and 23 released checkpoints on 12 datasets, finds cross-dataset rank correlations between detectors ranging from $\rho = 0.01$ to 0.87 and a 37-point spread between the best and worst method [8]. It also finds that training data explains 20–60% of the variance

among detectors of the same architecture, and that FLUX-dev, Firefly v4 and Midjourney v7 images are detected at only 18–30% on average [8]. A number measured on a curated benchmark is therefore an upper bound, not a deployment estimate.

B. Robustness has become the primary axis

Most robustness studies degrade images with parametric operators such as a JPEG quality factor or a Gaussian blur of chosen width. NTIRE 2026 moved closer to the wild. Its real and fake images are semantically matched and aligned in resolution, aspect ratio and JPEG quality, and its robust track applies one to five random consecutive distortions drawn from 36 types [10]. The winning systems show where the frontier is. They are ensembles of fine-tuned DINOv3 models. One team fine-tuned two 7B-parameter backbones on B200 GPUs and reports inference at 2.21 images/s using 78.25 GB of memory on one A100 [10]. That is far from the single-GPU budget of most deployments, which is why this paper studies what can be done with off-the-shelf models.

C. Shortcuts inflate reported numbers

A detector can separate real from fake without looking at generation artifacts at all, if the dataset lets it. Grommelt et al. show that GenImage's real images are mostly JPEG-compressed while its fakes are PNG, and that sizes differ systematically. Constraining training data to remove these biases raises cross-generator accuracy on GenImage from 71.68% to 82.74% for ResNet-50 and from 74.09% to 85.83% for Swin-T [9]. An open-source audit reports an extreme case in SID-Set [33]: all real images are JPEG and all fakes are identically sized PNG, so the rule "PNG means fake" scores 100% [34]. That audit is not peer-reviewed. Bias-free training addresses the problem from the training side [35]. For evaluation, the lesson is that containers must be normalized before any comparison, and that trivial baselines, such as file size at a fixed quality, belong in the results table.

A subtler shortcut is *resampling history*. If real photos are down-sampled to the evaluation size while fakes are generated at it, or real images are up-sampled from small web images, the resampling trace alone separates the classes. Our pilot (Section VII-B) therefore used real images whose short edge exceeds the target size, so that every real image is down-sampled and none is up-sampled.

D. Ranking is not deciding: calibration

Benchmarks favour threshold-free metrics. NTIRE's primary metric is ROC AUC, chosen precisely to avoid threshold effects [10]. A deployed system, however, must commit to a threshold, and the score distribution moves under distribution shift even when the ranking does not. The out-of-the-box audit uses a fixed 0.5 threshold and finds that AUC and accuracy correlate only at $r = 0.82$, with outliers that rank well but decide badly [8]. Recent work argues that calibration alone recovers much of the lost accuracy [36]. Our pilot makes the same point from the other side: ranking survives degradations that wreck a fixed threshold (Section VII-B).

V. Provenance and Watermarking

A. C2PA Content Credentials

The C2PA specification [4] defines a signed manifest embedded in the file. It records the tool that produced the asset, a list of actions, the ingredients it was derived from, and a signature by an X.509 certificate. A verifier checks the hash bindings to the pixels, the claim signature, and whether the signing certificate chains to a trust list. The actions can declare a digital source type from the IPTC vocabulary [37]. `trainedAlgorithmicMedia` means that a trained model produced the content, and `compositeWithTrainedAlgorithmicMedia` means it contains such elements. Two practical points are easy to get wrong.

First, `algorithmicMedia` without *trained* denotes content computed without a learned model, such as a fractal or a gradient. A detector that matches the substring "algorithmicMedia" will mislabel it as AI. Our own first implementation made this error. The official C2PA test fixture `C.jpg` declares exactly this source type.

Second, a manifest's presence proves little until it is verified. Anyone can sign a manifest with a self-issued certificate. Only a signature that chains to a trust list binds the claim to an accountable signer, and only intact hash bindings show that the pixels were not altered after signing.

B. Watermarks

Watermarks such as Tree-Ring [2] and SynthID [3] are embedded in the pixels at generation time and are designed to survive common transformations. They are strong evidence when present and a detector for them is available. A comparison of passive and watermark-based detection is given in [5].

C. Why provenance cannot replace passive detection

Both signals are one-sided. A verified manifest declaring AI, or a detected watermark, is close to conclusive. Their absence is not evidence of anything: social platforms strip metadata on upload, most generators do not sign, and open-weight models can be run without watermarking. A deployed tool should therefore report provenance *beside* the passive score, never fold it into the score, and never treat its absence as reassurance.

VI. Case Study: An Open Stateless Detector

A. Architecture

Fig. 1 shows the two tiers. A small FastAPI service on a VPS serves the page and the API, enforces an optional password and a per-IP rate limit, parses metadata and verifies C2PA, and forwards the image to a GPU function on a serverless platform that scales to zero. The GPU function runs the classifier on the whole image. It also runs the classifier on each cell of a 4×4 grid for a region heatmap, and computes three signal-processing diagnostics (high-frequency energy, noise-residual uniformity, JPEG 8-pixel grid discontinuity). The diagnostics are displayed for explanation only, with weight zero. The page warms a GPU container on load, so the 20–60 s cold start is usually hidden behind file selection.

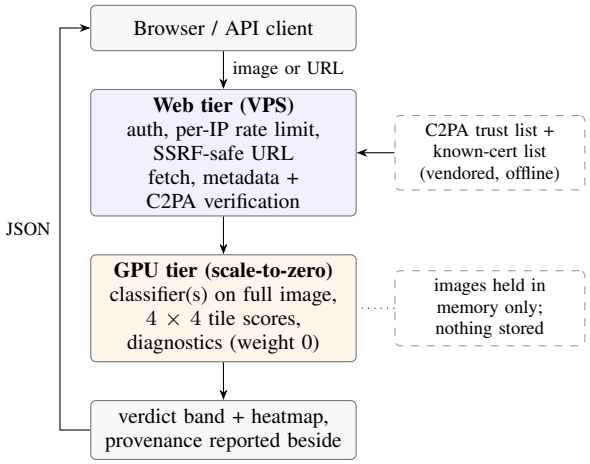


Fig. 1. System architecture. Images are held in memory only; no upload is stored. Provenance is computed on the web tier and reported beside the classifier score, never mixed into it.

B. Decisions driven by the review

Verdict bands, not a threshold. The deployed classifier is a Swinv2 model [38], [39]. On our pilot its optimal clean-image threshold was 0.994, and 20% of real photos scored above 0.5 (Section VII-B). The interface therefore uses three bands: "likely AI" (≥ 0.99), "inconclusive", and "no strong AI signal" (≤ 0.10). The meter uses a log-odds scale, because most scores concentrate within 1% of either end.

Heatmap from the same model. Every tile is scored independently by the classifier. A hot region inside a cool image suggests local editing; uniform heat is typical of fully generated images. The tiles are the model's own output, not a post-hoc saliency map.

Verified provenance. C2PA manifests are verified with the official library against the C2PA trust list and the known-certificate list used by the Content Credentials verifier, both vendored into the build. The interface separates four states: *trusted* (valid signature, listed signer), *valid* (intact but self-signed, so it proves nothing about the signer), *invalid* (altered after signing, or forged), and *absent*. An AI declaration anywhere in the edit history is reported, not only in the active manifest. Verification runs fully offline: remote manifests and OCSP are not fetched, so an upload cannot make the server issue requests.

URL input without SSRF. When the user supplies an image URL, the server resolves the host and requires every resolved address to be globally routable. It then connects to the checked address, with the hostname carried only in the Host header and TLS SNI, which defeats DNS rebinding. Every redirect hop is re-validated, and size and time are capped. The rate limit is keyed on the client address that the reverse proxy observed. Client-supplied forwarding headers are trusted only when they come from the proxy, and a CDN's client-IP header only when the proxy's peer lies in that CDN's published address ranges.

Stateless by design. No image is written to disk. This removes a class of privacy and retention obligations and keeps the service trivially replaceable.

VII. Evaluation

A. Protocol

All images, real and fake, were normalized to one container before any detector saw them: JPEG quality 95, 4:4:4 chroma, a 1024^2 centre square, sRGB, and no EXIF, ICC or XMP. Real images come from DIV2K [40], whose short edge (1,300–2,000 px) guarantees that every real image is down-sampled. Fakes were generated by FLUX.1-schnell [41] (natively 1024^2) and SDXL-base [42] at 1024^2 with 30 steps, from prompts spanning DIV2K's content categories. All detectors were run frozen and zero-shot on a single T4 GPU.

Pilot set. 25 real and 25 FLUX images under nine arms: clean; JPEG quality 90, 75 and 60; resize $\times 0.7$ and $\times 0.5$; Gaussian blur $\sigma = 0.5$; additive noise $\sigma = 2$; and resize$\times 0.7 \rightarrow$JPEG 75 stacked. This gives 450 paired cells. It is used for robustness analysis and for *all* threshold tuning.

Held-out set. The 75 DIV2K validation photographs not used in the pilot, plus 74 new FLUX and 74 SDXL images from 74 new prompts. No threshold or design choice was fitted on it.

Candidates for a second detector. From a screen of public models for licence, loadability and training data, we benchmarked two against the Swinv2 baseline. CommFor-384 is a ViT-S/16 [43] at 384 px, trained on Community Forensics' 4,803 generators [13], [44]. In that model, FLUX-schnell appears only in the evaluation split and DIV2K does not appear at all. The SigLIP model [45], [46] is a 224 px classifier with undisclosed training data. We combine scores by averaging log-odds ("mean"). We also report requiring both models to agree (the minimum probability) and a three-model mean.

B. Robustness of the baseline (pilot)

Three results from the pilot shaped the system. First, *ranking barely moved*. Swinv2's AUC ranged from 0.848 to 0.899 across all nine arms, and every 95% bootstrap interval overlapped. Second, *scores moved a lot*. Paired per-image shifts were large and highly significant: -0.169 under JPEG 60 (Wilcoxon $p = 8 \times 10^{-8}$), -0.089 under resize $\times 0.5$ ($p = 10^{-7}$), and $+0.123$ under noise ($p = 4 \times 10^{-7}$). At a 0.5 threshold, noise doubled the false-positive rate from 0.20 to 0.40, while compression drove it to 0 at the cost of recall. A study reporting only AUC would have concluded that nothing happened. Third, *a trivial baseline came close*. On clean images, logistic regression on file size alone reached AUC 0.844 against Swinv2's 0.883. Under noise, a baseline on three global pixel statistics beat Swinv2 outright (0.870 versus 0.848). We therefore report the file-size baseline in the held-out evaluation.

C. Second detector and ensemble (held-out)

Table III and Fig. 2 give the held-out results. The two single models are statistically indistinguishable overall (CommFor −

TABLE III
ROC AUC ON THE HELD-OUT SET (95% BOOTSTRAP CI). REAL: 75 DIV2K; FAKE: 74 FLUX.1-SCHNELL + 74 SDXL.

Detector	All	Real vs FLUX	Real vs SDXL
File size (baseline)	0.811 [.75,.87]	0.820	0.803
Swinv2	0.919 [.88,.95]	0.895	0.943
CommFor-384	0.925 [.89,.95]	0.853	0.996
SigLIP	0.873 [.83,.92]	0.906	0.839
Swinv2+CommFor mean	**0.956** [.93,.98]	0.916	0.997
Swinv2+CommFor min	0.950 [.92,.97]	0.909	0.991
All three, mean	0.966 [.94,.99]	0.940	0.992

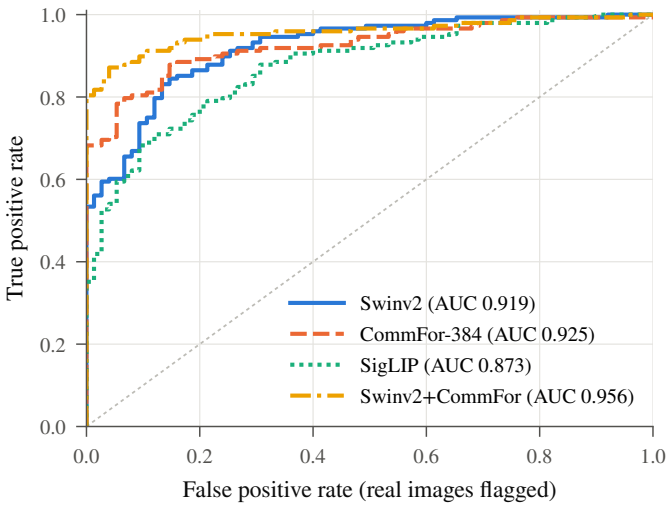


Fig. 2. ROC curves on the held-out set. The ensemble dominates at low false-positive rates, which is the regime a "likely AI" verdict operates in.

TABLE IV
OPERATING POINTS ON THE HELD-OUT SET. THE ENSEMBLE CUT-OFF WAS SET ON THE PILOT (REAL FPR $\leq 4\%$) AND APPLIED UNCHANGED. 95% WILSON INTERVALS.

Configuration	Reals flagged	Fakes caught
Swinv2 @ 0.99 (deployed)	6.7% (5/75) [2.9, 14.7]	65.5% (97/148) [57.6, 72.7]
Swinv2+CommFor @ 0.64	0.0% (0/75) [0.0, 4.9]	79.7% (118/148) [72.5, 85.4]

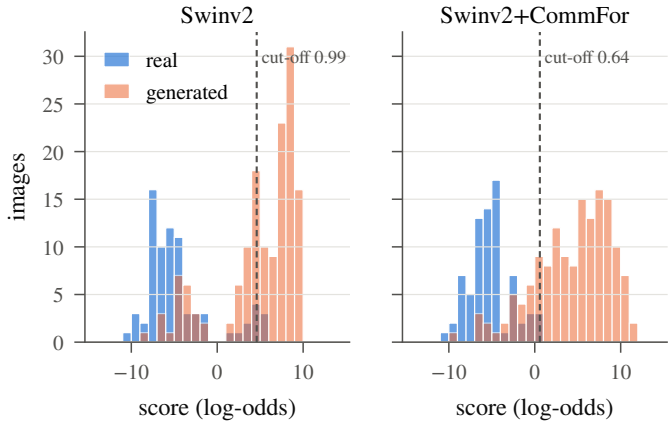


Fig. 3. Held-out score distributions in log-odds, with each configuration's "likely AI" cut-off. The ensemble separates the classes more cleanly and leaves no real photograph above its cut-off.

Swinv2: ΔAUC = +0.006, 95% CI [-0.039, +0.051]). They are, however, complementary: CommFor is near-perfect on SDXL and weaker on FLUX, and Swinv2 the reverse. Their score rank correlation is only $\rho = 0.62$. Averaging their log-odds gives a paired improvement over Swinv2 of ΔAUC = +0.037 (95% CI [+0.010, +0.068], bootstrap $P(\Delta \leq 0) = 0.004$), with both models evaluated on the same resampled images. Both models clear the file-size baseline by a wide margin, 0.108 for Swinv2 and 0.145 for the ensemble, a much larger gap than on the pilot.

What matters for a product is the operating point (Table IV, Fig. 3). At a cut-off chosen on the pilot, the ensemble flagged none of the 75 held-out real photographs and caught 79.7% of fakes. The deployed configuration flagged 6.7% of real photos and caught 65.5% of fakes. The false-positive intervals overlap, so the fair claim is a rate below roughly 5%, not zero. Calibration also transferred for the ensemble but not for CommFor alone. Tuned the same way, CommFor's false-positive rate rose from 4% on the pilot to 12% on held-out data.

Robustness trade-off. Under degradation (Table V, Fig. 4), CommFor alone is fragile: its AUC falls to 0.670 at JPEG 60 and 0.770 on the stacked arm. The ensemble inherits Swinv2's robustness. It is never below Swinv2 on any arm, equal at JPEG 60 and 0.015 to 0.053 higher elsewhere. At its conservative cut-off, it is more cautious than Swinv2 at 0.99 on degraded fakes, catching 24% versus 36% at JPEG 60. It is also more cautious on degraded *reals*, flagging 16% versus 24% under noise, the arm that most inflates false positives. For a tool where a false positive means accusing a person of fabrication, we consider this the right direction to err in.

D. A content-shortcut detector

The SigLIP model has the best pilot AUC in Table V, and its AUC does not change at all under any degradation. Per-image scores explain why: between an image and its JPEG-60 copy, its score changes by at most 0.007, and the mean absolute change is below 0.0005 on every arm. Swinv2 and CommFor move by 0.09–0.17 on average exactly where artifacts are destroyed. A detector that is invariant to compression, resizing and noise cannot be reading generation artifacts. It is almost certainly reading *content*, here the kinds of scenes our FLUX prompts produced. On held-out images from new prompts it fell to the worst AUC of the three (0.873), and to 0.839 on SDXL. Adding it to the ensemble raises held-out AUC to 0.966, but the gain rests on a shortcut that we expect to fail on new subject matter, so we exclude it. We suggest this invariance test as a cheap screen: *a detector whose scores do not move when artifacts are destroyed should not be trusted to have learned them.*

TABLE V
ROC AUC PER PILOT DEGRADATION ARM (25 REAL + 25 FLUX EACH).

	clean	J90	J75	J60	R.7	R.5	blur	noise	stack
Swinv2	.883	.899	.878	.880	.894	.891	.883	.848	.882
CommFor	.859	.856	.802	.670	.866	.870	.853	.851	.770
SigLIP	.930	.931	.931	.930	.931	.932	.933	.930	.930
Ensemble	.906	.914	.910	.880	.910	.922	.901	.901	.899

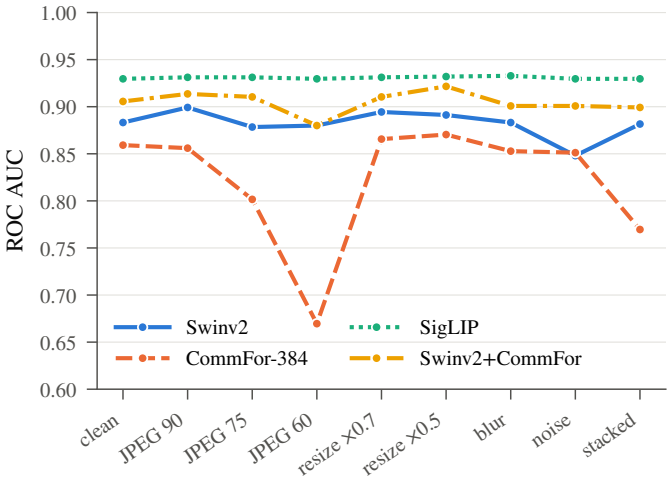


Fig. 4. AUC under synthetic degradation (pilot). CommFor collapses under strong JPEG; the ensemble never falls below Swinv2. SigLIP is flat, a warning sign discussed in Section VII-D.

VIII. DISCUSSION

A. Recommendations for practitioners

1) **Normalize containers and resampling history before evaluating.** Otherwise format, size or up-sampling can separate the classes on their own [9], [34].
2) **Put a trivial baseline in the adjacent column.** On our pilot, file size came within 0.04 AUC of a deployed detector.
3) **Report calibration beside AUC, and tune thresholds on separate data.** Ranking can be stable while the score distribution moves enough to double false positives.
4) **Prefer bands to a single threshold,** and make the middle band honest: "inconclusive" is a correct answer for much of the score range.
5) **Ensemble complementary detectors, not similar ones.** Low score correlation is the signal to look for. Combine in log-odds space and check each member's behaviour under degradation.
6) **Screen for content shortcuts** with the invariance test of Section VII-D.
7) **Verify provenance; don't sniff it.** Distinguish trusted, merely valid, and invalid manifests, and match IPTC source types exactly.
8) **Never present a score as proof.** Every detector evaluated here still misses a large share of fakes, and the literature reports far lower recall on recent commercial generators [8].

B. Open problems

Real redistribution pipelines. Synthetic operators are stand-ins. Messaging and social platforms apply undocumented, versioned resize and re-encode chains, and people screenshot and re-photograph screens. Whether detectors behave under these chains as they do under JPEG(q) is largely unmeasured. Our pilot suggests that the answer should be read in score shift and calibration, not only in AUC. **Moving targets.** Temporal benchmarks [14] and large generator pools [13] help, but any fixed detector will face generators newer than its training data. **Calibrated abstention.** Methods that know when they do not know, with guarantees under shift, matter more for deployment than another point of benchmark accuracy. **Faithful explanations.** VLM rationales [29], [30] need tests of whether the stated evidence is what drove the decision.

C. Limitations

Our evaluation is small: 75 real and 148 fake held-out images, one real source (professional photography), and two open generators. The intervals are honest but wide. DIV2K versus diffusion output is an easier contrast than phone photos versus diffusion output, which is also why the file-size baseline is strong, so our absolute numbers are upper bounds. SDXL is very likely inside CommFor's training distribution, which flatters its SDXL score. We tested no commercial generators, on which the literature reports much lower detection. Our robustness arms are synthetic, not platform round-trips.

IX. CONCLUSION

Passive detection of AI-generated images works well on curated benchmarks and much less well in the conditions where it is needed. The gap is driven by unseen generators, real-world degradation, dataset shortcuts, and thresholds that do not survive distribution shift. Our case study shows that careful evaluation can still produce a meaningfully better tool from public parts. Pairing two complementary off-the-shelf detectors gave a significant held-out AUC gain and, at a threshold fixed in advance, fewer false accusations with more fakes caught. The same evaluation also rejected a model that looked excellent on paper. The most useful habits are inexpensive: normalize containers, show a trivial baseline, report calibration, test invariance, verify provenance, and present scores as evidence rather than proof.

DATA AND CODE AVAILABILITY

Per-image scores, the evaluation harness, figure scripts and the benchmark report are included with the project repository.

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